Oscillatory neurobiomarkers for syntactic encoding during sentence planning

Author Information

Zhaoqian Yue[1†], Shuning Tang[2, 3†], Mingchuan Yang[4], Guang Yang[1], Peisong Yan[5, 6, 7], Dongwei Li[2, 3*], Angela D. Friederici[8*#], Luyao Chen[8, 9*]

1 School of International Chinese Language Education, Beijing Normal University, Beijing, China

2 Department of Psychology, Faculty of Arts and Sciences, Beijing Normal University, Zhuhai, China

3 Beijing Key Laboratory of Applied Experimental Psychology, National Demonstration Center for Experimental Psychology Education, Faculty of Psychology, Beijing Normal University, Beijing, China

4 Department of Linguistics and Modern Languages, The Chinese University of Hong Kong, Shatin, N.T., Hong Kong SAR, China

5 Department of Neurosurgery, Huashan Hospital, Shanghai Medical College, Fudan University, Shanghai, China

6 Shanghai Key Laboratory of Brain Function Restoration and Neural Regeneration, China

7 National Center for Neurological Disorders, Huashan Hospital, Shanghai Medical College, Fudan University, Shanghai 200040, China

8 Department of Language Science and Technology, Faculty of Humanities, Hong Kong Polytechnic University, Hong Kong, China

9 Department of Neuropsychology, Max Planck Institute for Human Cognitive and Brain Sciences, Leipzig, Germany

† These authors contributed equally to the present work as co-first authors

* Corresponding:

Dongwei Li: No. 18, Jinfeng Road, Tangjiawan, Zhuhai, China

Angela D. Friederici: Stephanstraße 1a, 04103 Leipzig, Germany

Luyao Chen: 8 Hung Lok Road, Hung Hom, Kowloon, Hong Kong, China

Email: DongweiLi@bnu.edu.cn (D. L.), friederici@cbs.mpg.de (A. D. F.), harry-luyao.chen@polyu.edu.hk (L. C.)

#Senior authorship

Abstract

Constructing well-structured sentences from preverbal messages is a core computational feat of human language production. Current proposals converge on the view that this structure-building is orchestrated by neural oscillations, yet the specific neural dynamics tracking the incremental generation of hierarchical structures during sentence planning leave a significant gap. Here, we employed a cued-delay picture description task to isolate syntactic computation from conceptual and articulatory confounds. Using scalp electroencephalography (EEG), multivariate pattern classification revealed that native speakers formed robust, sustained syntactic complexity representations. We then analyzed oscillatory dynamics at the scalp and source levels, as well as source-level functional connectivity, demonstrating that such representational stability was underpinned by theta oscillations and their phase synchronization across a specialized left-lateralized frontotemporal network. We further examined whether theta oscillations also serve as a sensitive index of syntactic encoding in second language (L2) learners and found that theta trajectories consistently distinguished the two groups: unlike the robust native pattern, L2 speakers exhibited delayed and transient syntactic complexity representations, recruiting a compensatory bilateral frontal-posterior network. Notably, theta activity flexibly tracked the subjective syntactic computational load specific to each group and significantly correlated with behavioral performance. Together, these results provide a multi-level account of sentence planning, establishing theta oscillations as critical neurobiomarkers that signal the temporal integration of syntactic hierarchical structure and index syntactic encoding efficiency across linguistic experiences.

Keywords

Sentence planning, Syntax, Neural dynamics, Theta oscillation, EEG

Introduction

As a remarkable "speaking animal," humans are significantly capable of generating well-structured sentences for communication. A key facet of understanding human language capacity involves elucidating how complex *internal languages* are generated and represented in our brains to convey various thoughts. Specifically, during sentence production, Levelt's model posits that preverbal messages are syntactically encoded to construct well-organized structures prior to articulation, which is at the core of the intermediate formulating phase (Bock & Levelt, 1994; Levelt, 1999). This psycho-cognitive model aligns well with the prominent linguistic model of the Minimalist Program framework, in which the core hierarchical syntactic computation (i.e., Merge, a binary syntactic operation combining two elements into a new constituent) plays a central role in internal language generation before externalization (Berwick et al., 2013; Chomsky, 2014). The neural implementation of such recursive computation necessitates the integration of two critical syntactic parameters: the 'vertical' relation, which distinguishes a higher-order constituent from its subordinate elements (e.g., differentiating a phrase from the words within it), and 'relational' information, which is essential for establishing dependencies between elements often spanning long distances (Coopmans et al., 2023). While emerging neuroimaging studies of sentence production focusing on spatial localization have increasingly highlighted that the left frontotemporal network serves as a pivotal circuit for hierarchical structure building (Giglio et al., 2022, 2024; Hu et al., 2023; Indefrey et al., 2001), it remains largely unknown how the brain orchestrates the dynamic neural planning for syntactic encoding with high temporal resolution, particularly regarding how relational dependencies between distant words are resolved prior to articulation.

A wide range of studies supports that sentence comprehension and production draw on the same linguistic knowledge representations, likely reflecting shared access to the core syntactic computation (Berwick et al., 2013; Friederici, 2017; Gambi & Pickering, 2017; Giglio et al., 2022, 2024; Hu et al., 2023; Pickering & Garrod, 2014). In sentence comprehension, neural oscillations, which have been shown to be substantial for reflecting the rhythmic activities of neural ensembles, have been proposed to play a key role in revealing the temporal dynamics of syntactic structure building (Buzsáki & Draguhn, 2004; Kazanina & Tavano, 2023). This neural oscillatory activity provides compelling evidence for elucidating the neural coding mechanisms underlying syntactic parsing (Coopmans et al., 2023; Ding, 2023; Kazanina & Tavano, 2023; Meyer et al., 2020a, 2020b). The computational demand of such parsing is directly modulated by sentence-inherent syntactic complexity, which typically arises from disruptions to canonical word order, particularly through syntactic movement operations that create long-distance filler-gap dependencies (Diessel, 2007; Gibson, 1998). In complex sentences with center-embedded relative clauses, the dependency length between a relative head and its trace differs across structural types; notably, unlike Indo-European languages (Diessel & Tomasello, 2005; Felser & Roberts, 2007; Gibson, 1998; Just et al., 1996; Santi & Grodzinsky, 2010), subject-relative clauses (SRC) typically exhibit longer dependency lengths than object-relative clauses (ORC) in Mandarin Chinese, thereby imposing higher syntactic complexity (Gibson & Wu, 2013; Hsiao & Gibson, 2003; Sung et al., 2016; Xu et al., 2024).

Processing such long-distance dependencies necessitates retaining the filler in working memory (WM) until the gap is encountered for successful integration. WM relies on the dynamic interaction between frontal executive and posterior representational systems, a process orchestrated by theta oscillations via gated processing windows to maintain specific informational content (Ratcliffe et al., 2022; Von Stein & Sarnthein, 2000). Crucially, evidence from sentence comprehension indicates that this oscillatory mechanism is recruited to support syntactic integration, where increased theta power directly indexes the elevated cognitive demand of maintaining fillers and retrieving embedded antecedents to resolve long-distance dependencies (Meyer et al., 2015; Xu et al., 2024). However, it remains unclear whether these oscillatory dynamics are similarly engaged in sentence production. Addressing this gap is essential for determining whether theta oscillations serve as a shared neural code across modalities. Furthermore, theoretical models and empirical studies suggest that phase coherence between communicating neuronal groups is a core mechanism supporting syntactic computation, dynamically integrating information via a non-isomorphic mapping between neural oscillatory activity and syntactic processing (Fries, 2005; Giraud & Poeppel, 2012; Kazanina & Tavano, 2023; Murphy, 2024). Building on this, we posit that theta oscillations may similarly index the dynamic neural planning of syntactic complexity during sentence production. We further hypothesize that the precise coordination between syntactic structural roles and their lexical fillers is supported by phase coherence between communicating neuronal groups, which dynamically integrates information across timescales and brain regions to establish hierarchical relationships prior to articulation.

Although emerging studies have begun to explore how language-specific syntactic properties modulate neural oscillatory activities during sentence planning, these investigations have primarily focused on languages with rich morphosyntactic systems (Egurtzegi et al., 2022; Sauppe et al., 2021). However, these conclusions are heavily predicated on the availability of overt morphological cues to guide structure building. It remains largely unknown whether similar oscillatory mechanisms govern sentence planning in isolating languages such as Mandarin Chinese, which lacks inflectional morphology and instead relies primarily on rigid word order and hierarchical nesting to encode syntactic relationships. To fill this gap, the present study aims to investigate the neural dynamics underlying syntactic encoding during sentence planning, with a particular focus on how neural oscillatory dynamics track the real-time construction of hierarchical structures that vary in dependency lengths.

To this end, we employed a cued-delay picture description paradigm, in which participants were required to generate the target sentences with center-embedded relative clauses in mind within the delay period before articulation, after observing pictures composed of geometric shapes of nouns and verbs to diminish concrete conceptual semantics as well as language experiences (see *Materials and Methods* for more details). In Study 1, we examined native Mandarin Chinese speakers to identify neural signatures of syntactic encoding during sentence planning. Multivariate pattern classification first established that syntactic complexity representations emerge and are sustained throughout sentence planning. We then characterized the underlying oscillatory dynamics at both scalp and source levels, combined with functional connectivity analysis, to determine whether theta oscillatory activities and their phase coherence constitute the neural substrate for syntactic encoding. Then, in Study 2, we tested whether such oscillations also serve as a sensitive neurobiomarker capable of differentiating syntactic processing profiles across linguistic experiences. We examined second language (L2) learners of Mandarin Chinese, predicting that if theta dynamics reliably index syntactic encoding efficiency, they should systematically distinguish first language (L1) from L2 sentence planning. In this work, we converged on the notion that theta rhythms appear to be a flexible neurobiomarker that scales with the distinct computational bottlenecks of native and non-native syntactic encoding during sentence planning.

Materials and Methods

Participants

Thirty-three right-handed native speakers of Mandarin Chinese (17 males, mean age = 21.24 ± 2.05 years) were recruited for Study 1, and thirty-two Mandarin Chinese L2 learners with Korean as their native language were recruited for Study 2 (16 males, mean age = 24.06 ± 2.38 years). Although Korean is a head-final SOV language whose typology differs from Indo-European languages, it exhibits similar processing advantages for SRC (Kwon et al., 2010, 2013; O'Grady et al., 2003). All participants had normal or corrected-to-normal vision, exhibited no color blindness or color weakness, and reported no reading difficulties or history of psychiatric or neurological disorders. They received monetary compensation and provided written informed consent.

All L2 participants were international students studying in mainland China during the experiment session, whose Mandarin Chinese proficiency had reached HSK (a standardized Chinese proficiency test, ranging from band 1 with low proficiency to 6 with advanced proficiency) band 5 or 6. Additionally, they completed a language background questionnaire, reporting that Mandarin Chinese was the language they were most familiar with after their native tongue, and that they began learning it at an average age of 16.16 ± 4.07 years. Furthermore, a rapid Mandarin Chinese cloze test was administered to all L2 participants to quantify their Mandarin Chinese proficiency (see Supplementary for more details; Feng et al., 2020).

Subjects were excluded based on the following criteria: three native and two L2 participants were excluded due to excessive head movements. Additionally, two native and three L2 participants were excluded for low task accuracy (i.e., errors exceeding 50% in each condition). Consequently, the final sample comprised twenty-eight native participants (14 males; mean age = 21.18 ± 2.07 years) and twenty-seven L2 participants (14 males; mean age = 24.07 ± 2.54 years). Ethical approval for the study was obtained from Beijing Normal University.

Materials

Participants were presented with pictures and asked to produce syntactically complex Mandarin Chinese sentences containing SRC and ORC, both of which involve reordering and integration across long-distance filler-gap dependencies (Caplan & Waters, 2001; Gibson, 1998, 2000), necessary for complex hierarchical syntactic processing (Fig. 1A). As illustrated in Fig 1B & 1C, to elicit these two sentence structures, participants viewed two-dimensional geometric shapes and action signs. This design could reduce semantic complexity, minimize the risk of naming ambiguities or conceptual confusion, and purify the syntactic encoding (Indefrey et al., 2001). These two sentence conditions were only minimally different in the order of the noun and verb in the embedded clauses.

A total of 120 images were assigned in a pseudo-random manner across six runs, with 10 SRC and 10 ORC per run. The shape order was pseudo-randomized to ensure that no more than two consecutive trials had the same sentence structure. For each trial, participants were instructed to produce sentences from left to right to minimize potential differences in visual input across conditions. They were required to describe the image in one sentence that included all three noun shapes and two actions, thus forming a complex sentence with a center-embedded relative clause. The action direction between the two shapes on the right was manipulated to create different sentences.

----- Insert Fig. 1 about here ----

Experimental procedure

Before the formal experiment, participants were briefed on the EEG recording procedures. Then, they completed a practice session comprising one picture-naming task and one sentence-production task. During EEG acquisition, the formal task was divided into six runs, each separated by a one-minute break, with EEG data recorded continuously throughout. Stimuli were presented through E-Prime version 3.0 (Psychology Software Tools, Inc., Pittsburgh, PA, USA). Each trial began with a fixation cross displayed for a variable duration of 1,000-1,500 ms, followed by a centrally presented image for 2,500 ms. After the image observation, a central fixation cross remained on the screen for 4,000 ms, during which participants formulated their responses. Participants were then given 6,000 ms to produce the sentence aloud as accurately as possible, following the onset of the "Go" signal. The entire experiment, including practice and rest intervals, lasted approximately 50 minutes per participant.

Data analysis

Behavioral data analysis

Sentence production was evaluated for accuracy and response time. A response was considered correct only if the prescribed sentence structure was produced. Incomplete responses or those containing errors, hesitations, repetitions, or self-corrections were coded as incorrect. For each sentence, the onset and offset time of its six constituent words were manually coded in Praat relative to the onset time of the "Go" signal, followed by a manual verification step. Three key temporal markers were derived (Fig. 1A): (1) T1on, sentence production start time, (2) T4off, the offset time of the fourth word, and (3) T6off, sentence duration. Critically, T4off served as a pivotal time point for comparing syntactic construction across sentence structures, marking the closure of the embedded relative clause.

To investigate the influence of syntactic complexity on sentence production, a linear mixed-effects model was used to analyze response time (Bates et al., 2015), and a generalized linear mixed-effects model was used to analyze accuracy (Jaeger, 2008). Both models were implemented via the lme4 package (version 1.1-35.5) in R (version 4.4.1) (Bates et al., 2015). Trials with incorrect responses were excluded from the response time analysis. We employed the maximal effect structure that allowed for convergence(Barr et al., 2013). Log-transformed response time and accuracy served as the respective dependent variables, with participants and items included as random effects. Specifically, in Study 1, the model included Condition as the fixed effect.

Subsequently, in Study 2, to examine group differences, the model included Condition, Group, and their interaction as fixed effects. Condition (SRC and ORC) and group (L1 and L2) were contrast-coded as follows: SRC = - 0.5, ORC = 0.5; L1 = - 0.5, L2 = 0.5.

EEG data acquisition and analysis

EEG recording and preprocessing

Participants were seated approximately 80 cm away from the computer screen in a soundproof, light-adjustable room. Continuous EEG data were recorded from 64 Ag/AgCl electrodes arranged according to the international 10–20 system using a SynAmps amplifier (BrainProducts, Inc.). Eye movements and blinks were monitored using an electrode placed 1 cm above the right eye to record the vertical electrooculogram (VEOG). Before recording, electrode impedances were maintained below 5 kΩ. EEG signals were referenced online to FCz, amplified with a bandpass filter from 0.01 to 1000 Hz, and digitized at 1000 Hz.

Data preprocessing was conducted using the EEGLAB toolbox (version 2023; Delorme & Makeig, 2004) and custom MATLAB (version 2023b) scripts. First, EEG signals were filtered with a bandpass of 0.5-40 Hz and an additional notch filter at 48-52 Hz to remove power-line interference. Signals were then re-referenced offline to the average electrodes. Data segments were extracted from -3500 ms to 4000 ms relative to the delay onset, with time windows defined as follows: baseline (-3500 to -2500 ms), image observation (-2500 to 0 ms), and delay (0 to 4000 ms). Incorrect trials were excluded from the analysis. Eyeblinks and movement artifacts were identified and removed using independent component analysis (ICA). Following ICA, baseline correction was applied to each segment. Additionally, segments with voltage exceeding ± 80 μV at any electrode were discarded. Participants were excluded from further analysis if more than 50% of their trials were rejected due to artifacts. On average, 89.58 ± 14.91 trials per participant were retained for further analysis. The number of trials per condition was as follows: 44.85 ± 6.92 for SRC and 44.73 ± 8.99 for ORC.

Multivariate pattern classification

The goal of multivariate pattern classification was to identify a linear combination of EEG electrode signals that could reliably distinguish between SRC and ORC. The time course of decoding performance, measured relative to chance, reveals whether and when brain activity encodes information about syntactic complexity during sentence planning. For each trial, the analysis time window spanned from -2700 ms to 4000 ms relative to the delay onset. To enhance decoding stability, we downsampled the data to 50 Hz by averaging adjacent time points (Li et al., 2023). At each time point, the classifier, based on a linear support vector machine (SVM), used responses from all electrodes as features. The classifiers were trained using the MATLAB *fitcecoc* function. For each data point, we applied 10-fold cross-validation and pseudo-averaged individual trials within each fold over 200 iterations to reduce variability introduced by trial assignments and ensure a more robust result (Ashton et al., 2022; Grootswagers et al., 2024). Decoding was considered accurate if the classifier correctly predicted SRC or ORC at a 50% chance level.

To examine the significance of decoding accuracy, segments were assigned random labels to train and test the control classifiers. Then, a 10,000-time cluster-based permutation test was performed across time (cluster-defining threshold $p < 0.05$; corrected $p < 0.05$; Bae & Luck, 2018).

Searchlight

To further elucidate the location information underlying the successful decoding of syntactic complexity representation, we employed a searchlight analysis, which is essentially a sliding window in the space of electrodes, performing classification within localized neighborhoods of electrodes rather than across all electrodes simultaneously (Bruera & Poesio, 2025; Graumann et al., 2022). By systematically shifting this window over the entire scalp, we can pinpoint which electrodes contribute most to distinguishing SRC from ORC, thereby offering insight into the spatiotemporal dynamics of syntactic encoding during sentence planning.

We conducted multivariate pattern classification as described above, with the following difference: for each electrode, we applied the classification procedure to the four closest electrodes surrounding it, based on Euclidean distance (Graumann et al., 2022). The decoding accuracy was then assigned to the central electrode, creating a topographical map of local decoding performance across the scalp.

Time-frequency analysis

To investigate the neural mechanisms underlying syntactic complexity representations during sentence planning, we performed time-frequency analysis of single-trial EEG data using a short-time Fourier transform (STFT) with the Hanning taper. This analysis covered a frequency range of 2-40 Hz, with a frequency resolution of 0.5 Hz, implemented via the *newtimef* function in EEGLAB. The power was then baseline-corrected between -700 and -300 ms prior to image observation and transformed to log space. The resulting power estimates were then averaged into three frequency bands:

theta (4-8 Hz), alpha (8-13 Hz), and beta (13-30 Hz) (Egurtzegi et al., 2022; Sauppe et al., 2021).

For statistical analysis, point-by-point paired t-tests were conducted at each electrode to compare power between the SRC and ORC conditions within the critical time windows identified by the decoding-behavior correlation analysis. To control for the family-wise error rate associated with multiple comparisons across all the electrodes, p-values were adjusted using the False Discovery Rate (FDR) correction method. Effects were considered significant at $q < 0.05$.

Theta source-level activation and functional connectivity analysis

To identify brain regions driving theta oscillations during syntactic encoding, we conducted source reconstruction focused on the theta band and targeted specific time windows based on sensor-level time-frequency findings.

We implemented a DICS beamformer approach (Gross et al., 2001) using Fieldtrip (Oostenveld et al., 2011). For each participant, we extracted baseline data and stimulus-related activity from time windows of interest. Cross-spectral density (CSD) matrices were calculated using the Hanning-tapered method with a target frequency of 6 (± 2) Hz. A standard single-shell head model derived from the ICBM152 template was used (Fonov et al., 2011), with lead fields constructed on a 0.8 cm resolution grid and normalized to MNI space. Source power was calculated in the Log space and interpolated onto the AAL90 atlas (Rolls et al., 2015). Statistical comparisons between SRC and ORC were performed using paired t-tests, with FDR correction for multiple comparisons across brain regions.

For functional connectivity analysis, we selected regions of interest based on theta source-level activation results to identify brain regions that could be functionally connected during syntactic encoding. Using a partial canonical coherence (PCC) beamformer, which extracts both power and phase information (Gross et al., 2001), we extracted phase information in source space. After concatenating SRC and ORC data, we calculated CSD matrices focused on the 6 (± 2) Hz frequency range. Source power and phase estimates were obtained from time windows of interest. The phase-locking value (PLV; Lachaux et al., 1999), which specifically captures phase synchrony independent of amplitude correlations, was computed between each dipole and averaged across the AAL90 atlas (Rolls et al., 2015), followed by a Fisher *Z* transformation.

Given the theta source-level results, we conducted a seed-based analysis using LIFG and bilateral temporoparietal regions to compare functional connectivity between SRC and ORC (Giglio et al., 2024). All reported effects have FDR-corrected *p*-values below 0.05.

Results

Study 1: Theta oscillations track syntactic complexity during native sentence planning

Neural dynamics of syntactic encoding predict behavioral performance. We first characterized the behavioral performance of native speakers (Fig. S1). For sentence accuracy, native speakers exhibited a marginally significant advantage for ORC structures ($b = 0.37$, SE = 0.19, $z = 1.90$, $p = 0.057$), a trend consistent with previous empirical findings (Chen et al., 2008; Gibson & Wu, 2013; Hsiao & Gibson, 2003; Sung et al., 2016; Xu et al., 2019). For response times, no significant differences were observed between these two structures.

To determine whether these behavioral patterns were underpinned by specific temporal dynamics of neural representations during sentence planning, multivariate pattern classification analysis was applied to identify whether different structures could be decoded from neural response distributions. Using right-tailed tests against the 50% chance level and applying the cluster-based permutation test, we identified a robust and sustained maintenance of syntactic complexity representations (Fig. 2A). Specifically, native speakers demonstrated significant classification of syntactic information throughout the delay period, with six significant clusters at 0-200 ms, 940-1140 ms, 1300-2160 ms, 2280-2500 ms, 2940-3400 ms, and 3660-3960 ms.

Next, to link these neural representations to planning efficiency, we correlated decoding accuracy with behavioral performance (Fig. 2B). We found that higher decoding accuracy during the early delay period (920-1200 ms) significantly predicted better sentence accuracy ($r = 0.530$, $p = 0.022$, FDR-corrected). Furthermore, searchlight analysis within this predictive window revealed that the informative patterns were spatially localized to the left anterior region ($p < 0.05$, FDR corrected; Fig. 2A).

Collectively, these results suggest that the ability to rapidly establish and sustain precise neural representations of syntactic structure is central to efficient sentence planning.

----- Insert Fig. 2 about here ----

Theta oscillations serve as a neural signature of syntactic encoding. Current proposals on the neural mechanisms of syntactic structure building converge on a key role for neural oscillations (Kazanina & Tavano, 2023). However, most evidence derives from sentence comprehension, with remarkably limited investigation into sentence production. To elucidate the oscillatory dynamics underlying syntactic complexity representation during sentence planning, we analyzed EEG power in different frequencies bands. Having identified the critical time window (920-1200 ms; Fig. 2B) where neural representations track behavioral performance, we sought to elucidate the specific oscillatory dynamics underlying this process. We found that syntactic complexity specifically modulated theta power rather than alpha or beta power (Fig 3). Specifically, planning for SRC structures elicited significantly greater theta power than planning for ORC structures. Point-by-point paired t-tests with FDR correction localized this effect to 8 significant electrodes (see Supplementary Table S1 for more statistical details). Notably, the significant effects exhibited clear spatial segregation, forming two focused clusters in the left anterior (F3, FC1, FC3, FC5; $p < 0.05$, FDR corrected) and right posterior (P4, CP6, P6; $p < 0.05$, FDR corrected). These findings demonstrate that increased syntactic complexity elicits stronger theta activity.

Functional coupling within a frontotemporal network. To pinpoint the neural generators underlying this theta modulation, we performed source localization (Fig. 4A). During the 1000-1200 ms interval after the delay onset, the increased theta activity for SRC was localized primarily to the right superior temporal gyrus (RSTG; $p = 0.041$, FDR corrected). Additional regions showing similar theta oscillatory patterns included the left inferior frontal gyrus (LIFG; uncorrected $p = 0.019$) and right supramarginal gyrus (RSMG; uncorrected $p = 0.032$).

Given the involvement of frontal and temporoparietal regions in syntactic encoding, we further examined theta-band functional connectivity between the LIFG and bilateral temporoparietal target regions. This analysis showed that connectivity between LIFG and LSTG was significantly stronger for SRC structures than for ORC structures ($t(27) = 3.876$, $p = 0.015$, Cohen's $d = 0.732$, FDR corrected).

Together, Study 1 establishes that focal, synchronized theta activity within a left-lateralized frontotemporal network constitutes the core neural mechanism supporting successful and efficient syntactic encoding during native sentence planning.

----- Insert Fig. 4 about here ----

Study 2: Theta oscillations serve as a sensitive neurobiomarker for differentiating linguistic experiences

Divergent behavioral patterns between L1 and L2 speakers. In Study 2, we aimed to validate whether the theta oscillatory mechanism identified in Study 1 could serve as a generalized neurobiomarker capable of distinguishing distinct syntactic encoding profiles associated with linguistic experiences during sentence planning. We recruited L2 learners of Mandarin Chinese to investigate variability in syntactic encoding efficiency. A generalized linear mixed-effects model on sentence accuracy revealed a significant main effect of Group ($b = -0.67$, SE = 0.16, $z = -4.15$, $p < 0.001$), indicating that sentence production performance was significantly better for native speakers. No significant main effect of Condition was observed ($b = 0.01$, SE = 0.14, $z = 0.08$, $p = 0.939$). Crucially, a significant Group × Condition interaction ($b = -0.69$, SE = 0.17, $z = -4.11$, $p < 0.001$) highlighted divergent syntactic preferences: native speakers performed better on ORC structures ($b = -0.36$, SE= 0.17, $z = 2.07$, $p = 0.039$), whereas L2 speakers showed a significant advantage for SRC structures ($b = 0.34$, SE= 0.16, $z = 2.16$, $p = 0.031$). Additionally, L2 speakers exhibited significantly longer planning times at the closure of the embedded clause (T4off: $b = 0.08$, SE= 0.01, $t = 5.79$, $p < 0.001$) and longer total sentence durations ($b = 0.09$, SE= 0.01, $t = 6.45$, $p < 0.001$), reflecting the higher cognitive demand of non-native syntactic encoding (Fig. S1).

Theta dynamics reveal compensatory syntactic encoding strategies. If theta oscillations are a robust neurobiomarker of syntactic encoding during sentence planning, they should reliably track these behavioral divergences. Indeed, multivariate pattern classification analysis for L2 speakers showed a markedly unstable classification pattern (Fig. 2A), with significant decoding clusters emerging only transiently (1160-1300 ms and 3480-3640 ms), suggesting an unstable maintenance of syntactic complexity representations.

We then investigated whether such complexity-dependent theta modulation represents a universal feature of syntactic encoding during sentence planning or is specific to the native neural architecture. We first analyzed the time window of interest defined by the decoding-behavior correlation (3420-3700 ms; $r = 0.468$, $p = 0.028$, FDR corrected; Fig. 2C). However, no condition-dependent modulation during this late interval was found, suggesting that while the successful syntactic complexity representation of L2 speakers was momentarily reactivated for articulation, the intense metabolic cost associated with complex structure building likely occurred at a different latency.

We therefore targeted the sustained planning phase (1000-3000 ms) to capture the underlying oscillation mechanisms. This exploratory analysis revealed a fundamentally different oscillatory pattern compared to that of native speakers (Fig. 3). While L2 speakers also exhibited theta modulation driven by sentence complexity, the oscillatory pattern was reversed and spatially widespread. In contrast to the native pattern, the statistical analysis identified a diffuse bilateral network of 27 significant electrodes, in which ORC planning elicited greater theta power ($p < 0.05$, FDR-corrected; see Table S2 for more statistical details). This reversal in power modulation and spatial diffusion suggests that theta oscillations flexibly scale with the specific computational bottlenecks of L2 processing, reflecting a compensatory mechanism to resolve the cognitive conflict inherent in complex structure building.

Brain-behavior coupling validates the theta neurobiomarker. Source localization of this compensatory theta activity (Fig. 4A) identified generators in the left middle temporal gyrus (LMTG; $p = 0.008$, FDR corrected) and the left inferior temporal gyrus (LITG; $p = 0.016$, FDR corrected). Theta modulation was also detected in the left inferior occipital gyrus (LIOG; uncorrected $p = 0.014$), LSTG (uncorrected $p = 0.032$), and RSMG (uncorrected $p = 0.015$).

Then, we validated the functional connectivity of these theta dynamics by phase coherence. In contrast to the native left-lateralized network, L2 speakers engaged a bilateral network, with significantly higher connectivity between the LIFG and bilateral SMG for the more difficult ORC structures (Fig. 4A). Most importantly, this connectivity strength predicted individual differences in processing efficiency. Significant negative correlations were found between LIFG-RSMG connectivity and both relative clause closure time (r = -0.397, p = 0.040, FDR corrected) and sentence duration (r = -0.397, p = 0.040, FDR corrected), indicating that faster production speeds in L2 participants were associated with phase-locking patterns more similar to those observed in native speakers (Fig. 4B).

Together, Study 2 confirms that theta oscillations are not merely a correlation of native processing but a sensitive neurobiomarker that dynamically indexes syntactic encoding efficiency across different linguistic experiences during sentence planning.

Discussion

Our study elucidates the real-time neural dynamics that support the planning of complex syntactic structures before speech. Using a cued-delay picture description paradigm that manipulated dependency length, we show that theta oscillations serve as a fundamental neural mechanism for orchestrating syntactic encoding during sentence planning. By integrating multivariate pattern analysis with scalp- and source-level oscillatory analyses, we established that theta rhythms function as a mechanistic "currency" that the brain allocates to maintain and access hierarchical syntactic structure until articulation, while flexibly tracking whichever operation constitutes the dominant computational bottleneck for a given speaker.

In native speakers, Study 1 establishes theta rhythms as a holding-and-readout mechanism that supports the maintenance of hierarchical structure over time. First, we observed that theta power was modulated by dependency length, consistent with previous findings linking theta power to syntactic building and retrieval in sentence comprehension (Bastiaansen et al., 2002, 2010; Meyer et al., 2015) and with EEG evidence from sentence planning showing theta activity is sensitive to morphosyntactic cues (Egurtzegi et al., 2022; Sauppe et al., 2021). Critically, our study extends these findings to the fundamental nature of sentence-inherent syntactic complexity. We reveal that theta rhythms specifically track the computational burden imposed by syntactic displacement operations (i.e., syntactic movement). By extending the linear distance between a filler and its gap, these operations increase the length of the relational dependency, thereby necessitating robust theta-mediated maintenance. For native speakers, the enhanced theta power during SRC planning indexes the elevated syntactic computation demand required to bridge this longer linear span (Gibson & Wu, 2013; Hsiao & Gibson, 2003; Sung et al., 2016; Xu et al., 2019), suggesting that in a highly automated syntactic processing system, the magnitude of theta power serves as a proxy for the computational effort of dependency formulation.

However, power alone cannot explain how precise structural relationships are maintained in real time. Given that hierarchical syntactic structure building relies on dynamic interactions across large-scale neural networks (Friederici, 2017; Giglio et al., 2024; Matchin & Hickok, 2019; Matchin & Wood, 2020), a precise temporal mechanism is required to coordinate these distributed representations of syntactic complexity. Therefore, we propose that theta phase dynamics serve as the essential holding mechanism that maintains the syntactic code. This interpretation is grounded in phase-coding models of WM, which posit that the theta cycle provides gated processing windows to separate and maintain individual informational items (Lisman & Jensen, 2013; Ratcliffe et al., 2022). Previous computational models further suggest that oscillations generate temporal phase codes to carry linguistic information forward in time, utilizing internal predictions to track structural dependencies (Ten Oever & Martin, 2021). Extending this to sentence planning, we posit that theta phase codes carry the identity of the filler forward, bridging the temporal gap to its trace. The robust and sustained decoding pattern we observed in native speakers indicates that this theta-mediated entrainment effectively prevents the structural prediction from decaying prior to articulation.

This phase-coded maintenance is not a passive storage state but a dynamic prerequisite for information readout. The functional necessity of this mechanism is corroborated by the tight temporal coupling we observed between the oscillatory and decoding patterns: the temporal window exhibiting the strongest theta-power modulation coincided precisely with the period during which neural patterns most accurately predicted behavioral performance. This convergence suggests that theta oscillations are mechanistically tied to the active access of syntactic information. Recent evidence demonstrates that frontal theta rhythms rhythmically scan WM representations, acting as a traveling wave that coordinates the precise timing of information readout (Han et al., 2026). Just as theta phase coordinates the spatial sampling of visual WM by sweeping across the representation, we posit that, in sentence planning, theta oscillations provide the temporal scaffolding for rhythmically scanning the hierarchical structure. This scanning mechanism keeps the long-distance filler active and accessible until it can be successfully integrated at the gap site. Thus, a unified mechanistic framework emerges: while theta power indexes the metabolic resources mobilized to meet syntactic computational demands, theta phase orchestrates the precise timing of the readout, ensuring the structural code is successfully linearized into speech.

Having established theta as a core neural mechanism for structural maintenance in native speakers, Study 2 aimed to assess its generalizability. Specifically, we asked: Does theta activity operate as a universal neurobiomarker for syntactic encoding during sentence planning, even when the underlying computational bottlenecks shift? Our results showed that L2 learners exhibited a reversed theta pattern, which precisely mirrored their behavioral difficulty. We argue that this divergence does not imply a different neural mechanism, but rather the same mechanism responding to a different computational constraint. For L2 planning, the primary burden plausibly shifts from sustaining long-distance dependencies to resolving conflict rooted in the Noun Phrase Accessibility Hierarchy (NPAH; Comrie & Keenan, 1979). Under transfer-driven, subject-first heuristics and accessibility constraints, constructing a non-canonical mapping (such as Mandarin Chinese ORCs for L2 learners with strong subject-first biases) may require inhibiting a prepotent strategy to assign the head noun to a less accessible role (Kwon et al., 2010; O'Grady et al., 2003; Ozeki & Shirai, 2007). The elevated theta power thus indexes the cognitive control required to resolve this thematic mapping conflict, overriding the structural subject bias to establish the non-canonical alignment (Cavanagh & Frank, 2014; Ness et al., 2024). Therefore, across groups, the theta mechanism emerges as a shared neurobiomarker of syntactic encoding that remains sensitive to the most computationally costly aspect of sentence planning.

Our network-level results further elucidate how these shared theta rhythms are spatially organized to support these distinct demands. While the source-level connectivity must be interpreted cautiously, given the spatial limitations of EEG (Hillebrand et al., 2012; Palva & Palva, 2012), native speakers showed a left-lateralized fronto-temporal pattern consistent with a specialized language circuit supporting efficient coordination between frontal control (Ratcliffe et al., 2022) and posterior lexical-syntactic representations (Friederici, 2011; Giglio et al., 2022, 2024; Matchin et al., 2019; Matchin & Wood, 2020). In contrast, L2 syntactic encoding engaged a bilateral network characterized by enhanced connectivity between the LIFG and bilateral posterior regions, which emerged considerably later during sentence planning. Rather than utilizing a specialized language network, these delayed bilateral interactions likely reflect a compensatory recruitment of domain-general cognitive control resources (Kormos, 2006; Wolna et al., 2024). This functional profile aligns with recent diffusion MRI evidence on structural plasticity in adult L2 learners, which demonstrates that language acquisition induces dynamic remodeling of bilateral white matter subnetworks and modulates interhemispheric connectivity (Wei et al., 2024). Specifically, as the specialized left-hemisphere network remains insufficient for high-demand L2 processing, the brain recruits the right hemisphere as a necessary scaffold to support lexical-syntactic integration. This difference is not at odds with the "same mechanism" account; instead, it suggests that the brain can instantiate theta-mediated syntactic encoding within distinct network architectures depending on expertise and processing constraints.

Notably, our results reveal an intriguing dissociation between the magnitude of theta oscillations and the robustness of syntactic representations. While L2 speakers exhibited robust, widely distributed theta-power differences across conditions, their decoding pattern remained transient and unstable. Conversely, native speakers showed more focal theta modulation but maintained a sustained syntactic complexity representation. This divergence likely reflects the distinction between cognitive effort and representational precision (Postle, 2021; Serences & Saproo, 2012). For L2 speakers, the widespread theta activation likely indexes a global recruitment of cognitive resources to resolve processing conflicts (e.g., inhibition of native transfer), a state of high metabolic consumption that does not necessarily equate to a precise neural code for the specific syntactic structure. In fact, such diffuse, compensatory activation may introduce neural noise, reducing the distinctiveness of the multivariate patterns required for successful classification (Haxby et al., 2014; Tong & Pratte, 2012). In contrast, the native pattern aligns with the neural efficiency hypothesis (Nussbaumer et al., 2015), which posits that expert performance is characterized by focused, energy-efficient neural engagement. The focal nature of native theta activity suggests a high signal-to-noise ratio in the neural circuit, allowing for the stable maintenance of distinct, fine-grained syntactic codes that are easily readable by the classifier despite lower overall global activation.

In summary, our study illuminates how humans temporally orchestrate the externalization of internal language. We demonstrate that the neural implementation of syntactic computation relies on frontal-posterior theta oscillations to resolve “relational” dependencies within “vertical” hierarchies. By leveraging this domain-general rhythmic scanning mechanism, the brain provides the essential temporal scaffolding to sequence hierarchical syntactic structures into linear speech. Whether resolving the relational dependencies of a native language or the mapping conflicts of a second language, this oscillatory orchestration stands as fundamental evidence of human syntactic encoding in sentence planning.

Declaration of competing interest

All authors approved the final version of the manuscript for submission and declared no conflict of interest.

Author contributions

Z. Y. and S. T.: Conceptualization, Data Curation, Formal Analysis, Methodology, Software, Visualization, and Writing-Original Draft. M. Y. and G. Y.: Methodology, Writing – review & editing, and Data curation. P. Y.: Methodology, and Writing – review & editing. A. D. F: Writing – review & editing. D. L. and L. C.: Conceptualization, Methodology, Supervision, Writing - review & editing, Funding acquisition.

Data availability

Anonymized data will be made available upon reasonable requests and upon a collaborative agreement addressed to the co-authors.

Acknowledgments

This work was supported by the STI 2030—Major Projects+2021ZD0200500, and the National Natural Science Foundation of China (32400863).

Reference

Ashton, K., Zinszer, B. D., Cichy, R. M., Nelson, C. A., Aslin, R. N., & Bayet, L. (2022). Time-resolved multivariate pattern analysis of infant EEG data: A practical tutorial. *Developmental Cognitive Neuroscience*, *54*, 101094. https://doi.org/10.1016/j.dcn.2022.101094

Bae, G.-Y., & Luck, S. J. (2018). Dissociable decoding of spatial attention and working memory from EEG oscillations and sustained potentials. *Journal of Neuroscience*, *38*(2), 409–422. https://doi.org/10.1523/JNEUROSCI.2860-17.2017

Barr, D. J., Levy, R., Scheepers, C., & Tily, H. J. (2013). Random effects structure for confirmatory hypothesis testing: Keep it maximal. *Journal of Memory and Language*, *68*(3), 255–278. https://doi.org/10.1016/j.jml.2012.11.001

Bastiaansen, M. C. M., Magyari, L., & Hagoort, P. (2010). Syntactic unification operations are reflected in oscillatory dynamics during on-line sentence comprehension. *Journal of Cognitive Neuroscience*, *22*(7), 1333–1347. https://doi.org/10.1162/jocn.2009.21283

Bastiaansen, M. C. M., Van Berkum, J. J. A., & Hagoort, P. (2002). Event-related theta power increases in the human EEG during online sentence processing. *Neuroscience Letters*, *323*(1), 13–16. https://doi.org/10.1016/S0304-3940(01)02535-6

Bates, D., Mächler, M., Bolker, B., & Walker, S. (2015). Fitting Linear Mixed-Effects Models Using lme4. *Journal of Statistical Software*, *67*(1). https://doi.org/10.18637/jss.v067.i01

Berwick, R. C., Friederici, A. D., Chomsky, N., & Bolhuis, J. J. (2013). Evolution, brain, and the nature of language. *Trends in Cognitive Sciences*, *17*(2), 89–98. https://doi.org/10.1016/j.tics.2012.12.002

Bock, K., & Levelt, W. J. M. (1994). Language production: Grammatical encoding. In M. A. Gernsbacher (Ed.), *Handbook of Psycholinguistics* (pp. 945–984). Academic Press.

Bruera, A., & Poesio, M. (2025). EEG searchlight decoding reveals person- and place-specific responses for semantic category and familiarity. *Journal of Cognitive Neuroscience*, *37*(1), 135–154. https://doi.org/10.1162/jocn_a_02125

Buzsáki, G., & Draguhn, A. (2004). Neuronal oscillations in cortical networks. *Science*, *304*(5679), 1926–1929. https://doi.org/10.1126/science.1099745

Caplan, D., & Waters, G. (2001). Working memory and syntactic processing in sentence comprehension. *Cognitive Studies: Bulletin of the Japanese Cognitive Science Society*, *8*(1), 10–24. https://doi.org/10.11225/jcss.8.10

Cavanagh, J. F., & Frank, M. J. (2014). Frontal theta as a mechanism for cognitive control. *Trends in Cognitive Sciences*, *18*(8), 414–421. https://doi.org/10.1016/j.tics.2014.04.012

Chen, B., Ning, A., Bi, H., & Dunlap, S. (2008). Chinese subject-relative clauses are more difficult to process than the object-relative clauses. *Acta Psychologica*, *129*(1), 61–65. https://doi.org/10.1016/j.actpsy.2008.04.005

Chomsky, N. (2014). *The Minimalist Program*. MIT Press.

Comrie, B., & Keenan, E. L. (1979). Noun phrase accessibility revisited. *Language*, *55*(3), 649–664. https://doi.org/10.2307/413321

Coopmans, C. W., Mai, A., Slaats, S., Weissbart, H., & Martin, A. E. (2023). What oscillations can do for syntax depends on your theory of structure building. *Nature Reviews Neuroscience*, *24*(11), 723–723. https://doi.org/10.1038/s41583-023-00734-5

Delorme, A., & Makeig, S. (2004). EEGLAB: An open source toolbox for analysis of single-trial EEG dynamics including independent component analysis. *Journal of Neuroscience Methods*, *134*(1), 9–21. https://doi.org/10.1016/j.jneumeth.2003.10.009

Diessel, H. (2007). Frequency effects in language acquisition, language use, and diachronic change. *New Ideas in Psychology*, *25*(2), 108–127. https://doi.org/10.1016/j.newideapsych.2007.02.002

Diessel, H., & Tomasello, M. (2005). A new look at the acquisition of relative clauses. *Language*, *81*(4), 882–906. https://doi.org/10.1353/lan.2005.0169

Ding, N. (2023). Low-frequency neural parsing of hierarchical linguistic structures. *Nature Reviews Neuroscience*, *24*(12), 792–792. https://doi.org/10.1038/s41583-023-00749-y

Egurtzegi, A., Blasi, D. E., Bornkessel-Schlesewsky, I., Laka, I., Meyer, M., Bickel, B., & Sauppe, S. (2022). Cross-linguistic differences in case marking shape neural power dynamics and gaze behavior during sentence planning. *Brain and Language*, *230*, 105127. https://doi.org/10.1016/j.bandl.2022.105127

Felser, C., & Roberts, L. (2007). Processing *wh* -dependencies in a second language: A cross-modal priming study. *Second Language Research*, *23*(1), 9–36. https://doi.org/10.1177/0267658307071600

Feng L., Feng H., Bai S., & Wu J. (2020). An Analysis of Proficiency Test for CSL (Chinese as Second Language): Based on Fixed-Ratio Cloze Questions. *Applied Linguistics*, (3), 69–79. https://doi.org/10.16499/j.cnki.1003-5397.2020.03.011

Fonov, V., Evans, A. C., Botteron, K., Almli, C. R., McKinstry, R. C., & Collins, D. L. (2011). Unbiased average age-appropriate atlases for pediatric studies. *Neuroimage*, *54*(1), 313–327. https://doi.org/10.1016/j.neuroimage.2010.07.033

Friederici, A. D. (2011). The brain basis of language processing: From structure to function. *Physiological Reviews*, *91*(4), 1357–1392. https://doi.org/10.1152/physrev.00006.2011

Friederici, A. D. (2017). *Language in our brain: The origins of a uniquely human capacity*. MIT Press.

Fries, P. (2005). A mechanism for cognitive dynamics: Neuronal communication through neuronal coherence. *Trends in Cognitive Sciences*, *9*(10), 474–480. https://doi.org/10.1016/j.tics.2005.08.011

Gambi, C., & Pickering, M. J. (2017). Models linking production and comprehension. In E. M. Fernández & H. S. Cairns (Eds.), *The Handbook of Psycholinguistics* (1st ed., pp. 157–181). Wiley. https://doi.org/10.1002/9781118829516.ch7

Gibson, E. (1998). Linguistic complexity: Locality of syntactic dependencies. *Cognition*, *68*(1), 1–76. https://doi.org/10.1016/S0010-0277(98)00034-1

Gibson, E. (2000). The dependency locality theory: A distance-based theory of linguistic complexity. In *Image, language, brain* (pp. 95–126). https://www2.bcs.rochester.edu/sites/fjaeger/teaching/LabSyntax2006/readings/Gibson_2000.pdf

Gibson, E., & Wu, H.-H. I. (2013). Processing Chinese relative clauses in context. *Language and Cognitive Processes*, *28*(1–2), 125–155. https://doi.org/10.1080/01690965.2010.536656

Giglio, L., Ostarek, M., Weber, K., & Hagoort, P. (2022). Commonalities and asymmetries in the neurobiological infrastructure for language production and comprehension. *Cerebral Cortex*, *32*(7), 1405–1418. https://doi.org/10.1093/cercor/bhab287

Giglio, L., Sharoh, D., Ostarek, M., & Hagoort, P. (2024). Connectivity of fronto-temporal regions in syntactic structure building during speaking and listening. *Neurobiology of Language*, *5*(4), 922–941. https://doi.org/10.1162/nol_a_00154

Giraud, A.-L., & Poeppel, D. (2012). Cortical oscillations and speech processing: Emerging computational principles and operations. *Nature Neuroscience*, *15*(4), 511–517. https://doi.org/10.1038/nn.3063

Graumann, M., Ciuffi, C., Dwivedi, K., Roig, G., & Cichy, R. M. (2022). The spatiotemporal neural dynamics of object location representations in the human brain. *Nature Human Behaviour*, *6*(6), 796–811. https://doi.org/10.1038/s41562-022-01302-0

Grootswagers, T., Robinson, A. K., Shatek, S. M., & Carlson, T. A. (2024). Mapping the dynamics of visual feature coding: Insights into perception and integration. *PLOS Computational Biology*, *20*(1), e1011760. https://doi.org/10.1371/journal.pcbi.1011760

Gross, J., Kujala, J., Hämäläinen, M., Timmermann, L., Schnitzler, A., & Salmelin, R. (2001). Dynamic imaging of coherent sources: Studying neural interactions in the human brain. *Proceedings of the National Academy of Sciences*, *98*(2), 694–699. https://doi.org/10.1073/pnas.98.2.694

Han, H.-B., Brincat, S. L., Buschman, T. J., & Miller, E. K. (2026). Working memory readout varies with frontal theta rhythms. *Neuron*, *114*(1), 159-166.e2. https://doi.org/10.1016/j.neuron.2025.09.031

Haxby, J. V., Connolly, A. C., & Guntupalli, J. S. (2014). Decoding neural representational spaces using multivariate pattern analysis. *Annual Review of Neuroscience*, *37*(1), 435–456. https://doi.org/10.1146/annurev-neuro-062012-170325

Hillebrand, A., Barnes, G. R., Bosboom, J. L., Berendse, H. W., & Stam, C. J. (2012). Frequency-dependent functional connectivity within resting-state networks: An atlas-based MEG beamformer solution. *Neuroimage*, *59*(4), 3909–3921. https://doi.org/10.1016/j.neuroimage.2011.11.005

Hsiao, F., & Gibson, E. (2003). Processing relative clauses in Chinese. *Cognition*, *90*(1), 3–27. https://doi.org/10.1016/S0010-0277(03)00124-0

Hu, J., Small, H., Kean, H., Takahashi, A., Zekelman, L., Kleinman, D., Ryan, E., Nieto-Castañón, A., Ferreira, V., & Fedorenko, E. (2023). Precision fMRI reveals that the language-selective network supports both phrase-structure building and lexical access during language production. *Cerebral Cortex*, *33*(8), 4384–4404. https://doi.org/10.1093/cercor/bhac350

Indefrey, P., Brown, C. M., Hellwig, F., Amunts, K., Herzog, H., Seitz, R. J., & Hagoort, P. (2001). A neural correlate of syntactic encoding during speech production.

Proceedings of the National Academy of Sciences, *98*(10), 5933–5936. https://doi.org/10.1073/pnas.101118098

Jaeger, T. F. (2008). Categorical data analysis: Away from ANOVAs (transformation or not) and towards logit mixed models. *Journal of Memory and Language*, *59*(4), 434–446. https://doi.org/10.1016/j.jml.2007.11.007

Just, M. A., Carpenter, P. A., Keller, T. A., Eddy, W. F., & Thulborn, K. R. (1996). Brain activation modulated by sentence comprehension. *Science*, *274*(5284), 114–116. https://doi.org/10.1126/science.274.5284.114

Kazanina, N., & Tavano, A. (2023). What neural oscillations can and cannot do for syntactic structure building. *Nature Reviews Neuroscience*, *24*(2), 113–128. https://doi.org/10.1038/s41583-022-00659-5

Kormos, J. (2006). *Speech Production and Second Language Acquisition*. Routledge. https://www.lightgoldm.com/speech%20production.pdf

Kwon, N., Gordon, P. C., Lee, Y., Kluender, R., & Polinsky, M. (2010). Cognitive and linguistic factors affecting subject/object asymmetry: An eye-tracking study of prenominal relative clauses in korean. *Language*, *86*(3), 546–582. https://doi.org/10.1353/lan.2010.0006

Kwon, N., Kluender, R., Kutas, M., & Polinsky, M. (2013). Subject/object processing asymmetries in Korean relative clauses: Evidence from ERP data. *Language*, *89*(3), 537–585. https://doi.org/dx.doi.org/10.1353/lan.2013.0044

Lachaux, J.-P., Rodriguez, E., Martinerie, J., & Varela, F. J. (1999). Measuring phase synchrony in brain signals. *Human Brain Mapping*, *8*(4), 194–208.

https://doi.org/10.1002/(SICI)1097-0193(1999)8:4%253C194::AID-HBM4%253E3.0.CO;2-C

Levelt, W. J. M. (1999). Producing spoken language: A blueprint of the speaker. In C. M. Brown & P. Hagoort (Eds.), *Neurocognition of language* (pp. 83–122). Oxford University Press. https://doi.org/10.1093/acprof:oso/9780198507932.003.0004

Li, D., Hu, Y., Qi, M., Zhao, C., Jensen, O., Huang, J., & Song, Y. (2023). Prioritizing flexible working memory representations through retrospective attentional strengthening. *NeuroImage*, *269*, 119902. https://doi.org/10.1016/j.neuroimage.2023.119902

Lisman, J. E., & Jensen, O. (2013). The Theta-Gamma Neural Code. *Neuron*, *77*(6), 1002–1016. https://doi.org/10.1016/j.neuron.2013.03.007

Matchin, W., Brodbeck, C., Hammerly, C., & Lau, E. (2019). The temporal dynamics of structure and content in sentence comprehension: Evidence from fMRI-constrained MEG. *Human Brain Mapping*, *40*(2), 663–678. https://doi.org/10.1002/hbm.24403

Matchin, W., & Hickok, G. (2019). The cortical organization of syntax. *Cerebral Cortex*, *30*(3), 1481–1498. https://doi.org/doi:%252010.1093/cercor/bhz180

Matchin, W., & Wood, E. (2020). Syntax-sensitive regions of the posterior inferior frontal gyrus and the posterior temporal lobe are differentially recruited by production and perception. *Cerebral Cortex Communications*, *1*(1), tgaa029. https://doi.org/10.1093/texcom/tgaa029

Meyer, L., Grigutsch, M., Schmuck, N., Gaston, P., & Friederici, A. D. (2015). Frontal–posterior theta oscillations reflect memory retrieval during sentence comprehension. *Cortex*, *71*, 205–218. https://doi.org/10.1016/j.cortex.2015.06.027

Meyer, L., Sun, Y., & Martin, A. E. (2020a). “Entraining” to speech, generating language? *Language, Cognition and Neuroscience*, *35*(9), 1138–1148. https://doi.org/10.1080/23273798.2020.1827155

Meyer, L., Sun, Y., & Martin, A. E. (2020b). Synchronous, but not entrained: Exogenous and endogenous cortical rhythms of speech and language processing. *Language, Cognition and Neuroscience*, *35*(9), 1089–1099. https://doi.org/10.1080/23273798.2019.1693050

Murphy, E. (2024). ROSE: A neurocomputational architecture for syntax. *Journal of Neurolinguistics*, *70*, 101180. https://doi.org/10.1016/j.jneuroling.2023.101180

Ness, T., Langlois, V. J., Novick, J. M., & Kim, A. E. (2024). Theta-band neural oscillations reflect cognitive control during language processing. *Journal of Experimental Psychology: General*, *153*(9), 2279–2298. https://doi.org/10.1037/xge0001621

Nussbaumer, D., Grabner, R. H., & Stern, E. (2015). Neural efficiency in working memory tasks: The impact of task demand. *Intelligence*, *50*, 196–208. https://doi.org/10.1016/j.intell.2015.04.004

O’Grady, W., Lee, M., & Choo, M. (2003). A subject-object asymmetry in the acquisition of relative clauses in korean as a second language. *Studies in Second Language Acquisition*, *25*(3), 433–448. https://doi.org/10.1017/S0272263103000172

Oostenveld, R., Fries, P., Maris, E., & Schoffelen, J.-M. (2011). FieldTrip: Open source software for advanced analysis of MEG, EEG, and invasive electrophysiological data. *Computational Intelligence and Neuroscience*, *2011*, 1–9. https://doi.org/10.1155/2011/156869

Ozeki, H., & Shirai, Y. (2007). Does the noun phrase accessibility hierarchy predict the difficulty order in the acquisition of japanese relative clauses? *Studies in Second Language Acquisition*, *29*(02). https://doi.org/10.1017/S0272263107070106

Palva, S., & Palva, J. M. (2012). Discovering oscillatory interaction networks with M/EEG: Challenges and breakthroughs. *Trends in Cognitive Sciences*, *16*(4), 219–230. https://doi.org/10.1016/j.tics.2012.02.004

Pickering, M. J., & Garrod, S. (2014). Neural integration of language production and comprehension. *Proceedings of the National Academy of Sciences*, *111*(43), 15291–15292. https://doi.org/10.1073/pnas.1417917111

Postle, B. R. (2021). Cognitive neuroscience of visual working memory. In R. Logie, V. Camos, & N. Cowan (Eds.), *Working memory* (pp. 333–343). Oxford University Press.

Ratcliffe, O., Shapiro, K., & Staresina, B. P. (2022). Fronto-medial theta coordinates posterior maintenance of working memory content. *Current Biology*, *32*(10), 2121-2129.e3. https://doi.org/10.1016/j.cub.2022.03.045

Rolls, E. T., Joliot, M., & Tzourio-Mazoyer, N. (2015). Implementation of a new parcellation of the orbitofrontal cortex in the automated anatomical labeling atlas. *Neuroimage*, *122*, 1–5. https://doi.org/10.1016/j.neuroimage.2015.07.075

Santi, A., & Grodzinsky, Y. (2010). fMRI adaptation dissociates syntactic complexity dimensions. *Neuroimage*, *51*(4), 1285–1293. https://doi.org/10.1016/j.neuroimage.2010.03.034

Sauppe, S., Choudhary, K. K., Giroud, N., Blasi, D. E., Norcliffe, E., Bhattamishra, S., Gulati, M., Egurtzegi, A., Bornkessel-Schlesewsky, I., Meyer, M., & Bickel, B. (2021). Neural signatures of syntactic variation in speech planning. *PLOS Biology*, *19*(1), e3001038. https://doi.org/10.1371/journal.pbio.3001038

Serences, J. T., & Saproo, S. (2012). Computational advances towards linking BOLD and behavior. *Neuropsychologia*, *50*(4), 435–446. https://doi.org/10.1016/j.neuropsychologia.2011.07.013

Sung, Y.-T., Tu, J.-Y., Cha, J.-H., & Wu, M.-D. (2016). Processing preference toward object-extracted relative clauses in mandarin chinese by L1 and L2 speakers: An eye-tracking study. *Frontiers in Psychology*, *7*. https://doi.org/10.3389/fpsyg.2016.00004

Ten Oever, S., & Martin, A. E. (2021). An oscillating computational model can track pseudo-rhythmic speech by using linguistic predictions. *Elife*, *10*. https://doi.org/10.7554/elife.68066

Tong, F., & Pratte, M. S. (2012). Decoding patterns of human brain activity. *Annual Review of Psychology*, *63*, 483–509. https://doi.org/10.1146/annurev-psych-120710-100412

Von Stein, A., & Sarnthein, J. (2000). Different frequencies for different scales of cortical integration: From local gamma to long range alpha/theta synchronization.

International Journal of Psychophysiology, *38*(3), 301–313. https://doi.org/10.1016/S0167-8760(00)00172-0

Wei, X., Gunter, T. C., Adamson, H., Schwendemann, M., Friederici, A. D., Goucha, T., & Anwander, A. (2024). White matter plasticity during second language learning within and across hemispheres. *Proceedings of the National Academy of Sciences*, *121*(2), e2306286121. https://doi.org/10.1073/pnas.2306286121

Wolna, A., Szewczyk, J., Diaz, M., Domagalik, A., Szwed, M., & Wodniecka, Z. (2024). Domain-general and language-specific contributions to speech production in a second language: An fMRI study using functional localizers. *Scientific Reports*, *14*(1), 57. https://doi.org/10.1038/s41598-023-49375-9

Xu, K., Duann, J.-R., Hung, D. L., & Wu, D. H. (2019). Preference for object relative clauses in Chinese sentence comprehension: Evidence from online self-paced reading time. *Frontiers in Psychology*, *10*, 2210. https://doi.org/10.3389/fpsyg.2019.02210

Xu, K., Ma, C., Liu, Y., & Duann, J.-R. (2024). Event-related potentials and brain oscillations reflect unbalanced allocation of retrieval and integration efforts in sentence comprehension. *Language, Cognition and Neuroscience*, *39*(2), 232–250. https://doi.org/10.1080/23273798.2023.2263582

Figures

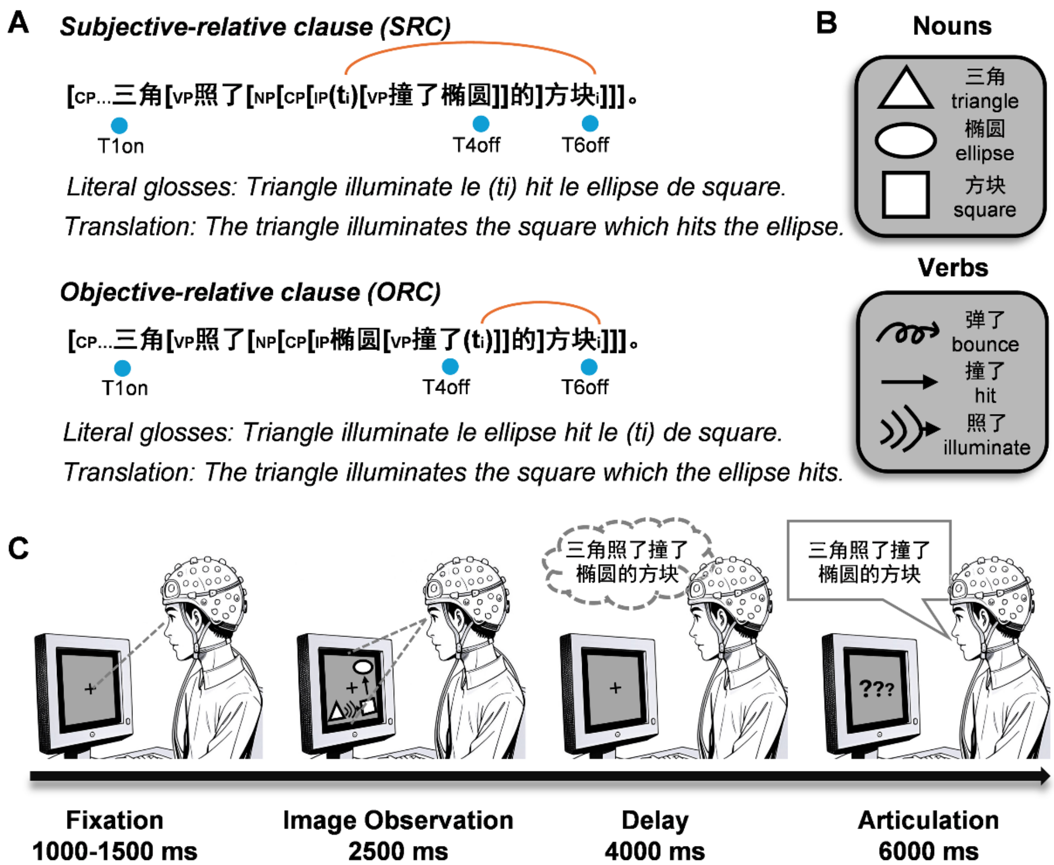

Fig. 1. Stimulus presentation. (A) Examples of SRC and ORC used in the Mandarin Chinese sentence production task (Literal glosses and English translations in italics). Three critical temporal markers were designated for subsequent analyses: T1on (sentence production onset), T4off (closure of the relative clause), and T6off (sentence duration). As illustrated, in SRC, a verb phrase (VP) is center-embedded between the trace (t) and its corresponding noun (N), co-indexed by the subscript "i", resulting in greater structural complexity (dependency between t_i and N_i indicated by the orange arc). In contrast, the ORC exhibits a simpler embedding structure, with the dependency between the trace (t_i) and the noun (N_i) similarly indicated by the orange arc. (B) Components that make up the visual stimuli. (C) Sample trials of the experiment.

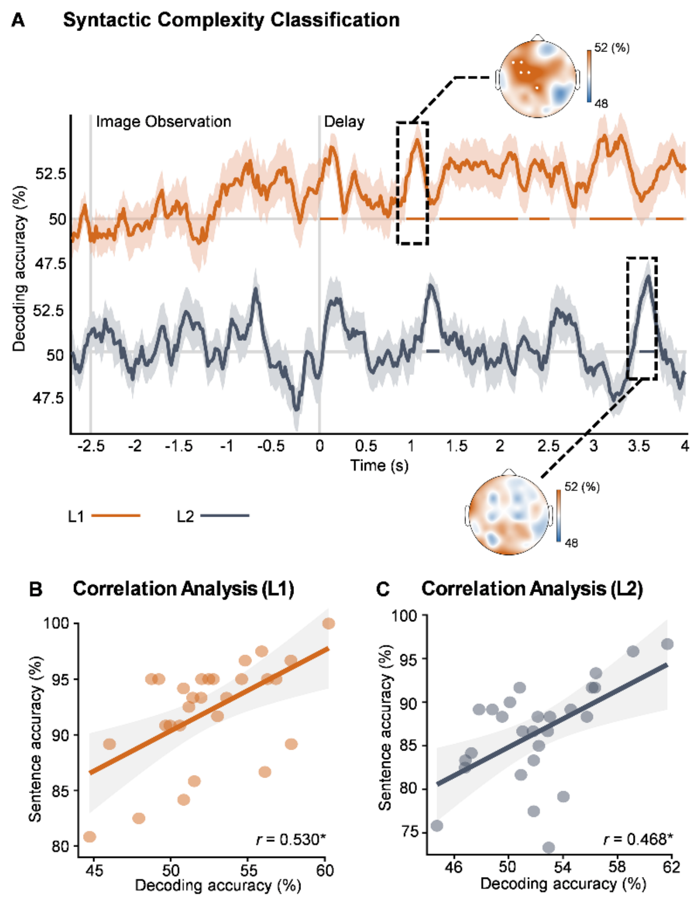

Fig. 2. (A) Temporal dynamics of decoding accuracy for L1 (orange) and L2 (grey) speakers. Solid horizontal lines indicate time windows with significantly above-chance decoding accuracy (cluster-based permutation test, $p < 0.05$). Topographical maps (insets) show the spatial searchlight analysis results during these significant time windows, with significant electrodes (F3, F5, FC1, FC3, CPz) marked by white dots ($p <$ 0.05, right-tailed t-tests, FDR corrected). (B) & (C) Correlations between decoding accuracy (averaged over 920-1200 ms for L1; 3420-3700 ms for L2) and mean behavioral accuracy (averaged across SRC and ORC conditions). * $p < 0.05$; *** $p <$ 0.001.

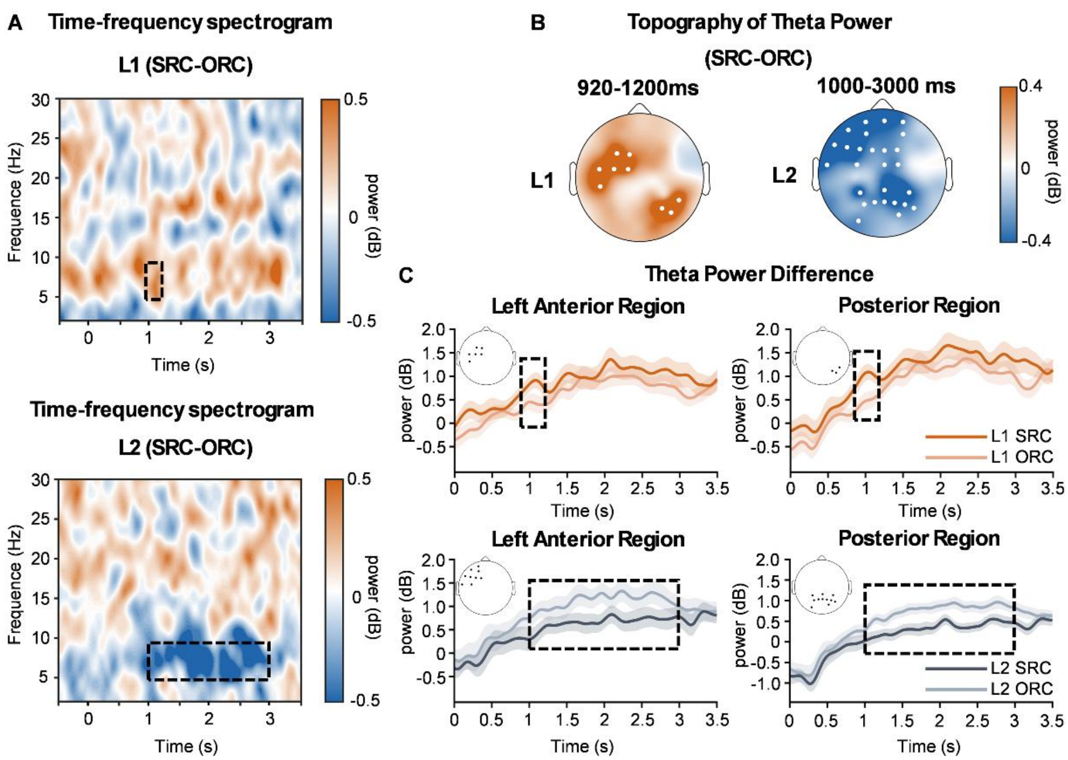

Fig. 3. Scalp-level oscillatory results. (A) Time-frequency spectrograms of the power difference between SRC and ORC conditions (SRC minus ORC), averaged across all electrodes. For L1 speakers (top), SRC elicited significantly higher theta power than ORC, particularly within the 920-1200 ms window (highlighted by the black dashed rectangle). Conversely, L2 speakers (bottom) showed reduced theta power for SRC relative to ORC within the 1000-3000 ms window. (B) Topographical maps illustrate theta-band power differences (SRC minus ORC) within the significant time windows identified in (A). White dots indicate electrodes with significant differences ($p < 0.05$, two-tailed t-tests, FDR corrected). (C) Time courses of mean theta power for SRC and ORC conditions extracted from significant electrode clusters in the left anterior and posterior regions. Shaded areas represent the standard error of the mean. Black dashed rectangles denote time intervals with significant effects.

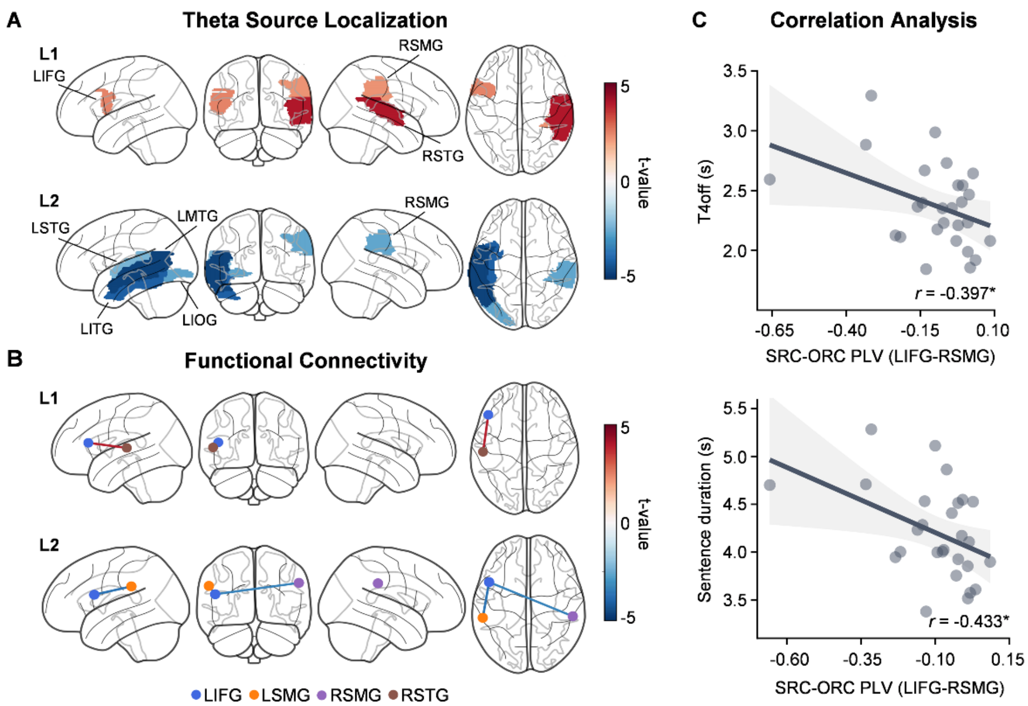

Fig. 4. Source-level theta activity activation and functional connectivity. (A) Cortical distribution of *t*-values representing the contrast in theta power between SRC and ORC conditions (masked at $p < 0.05$). L1 speakers (top, 1000-1200 ms) show increased theta activity in frontotemporal regions, while L2 speakers (bottom, 1800-2200 ms) exhibit reduced theta activity in temporoparietal regions. (B) Functional connectivity analysis showing *t*-values for theta phase synchrony differences between LIFG and bilateral temporoparietal regions (masked at $p < 0.05$, FDR corrected). Red lines indicate enhanced connectivity (L1), and blue lines indicate reduced connectivity (L2). (C) Scatter plots illustrating relationships between connectivity and behavioral performance. A significant negative correlation was found between the modulation of LIFG-RSMG connectivity (SRC minus ORC) and response times (T4 offset and sentence duration), indicating that stronger functional coupling is associated with faster sentence processing. * $p < 0.05$.

Supplementary information

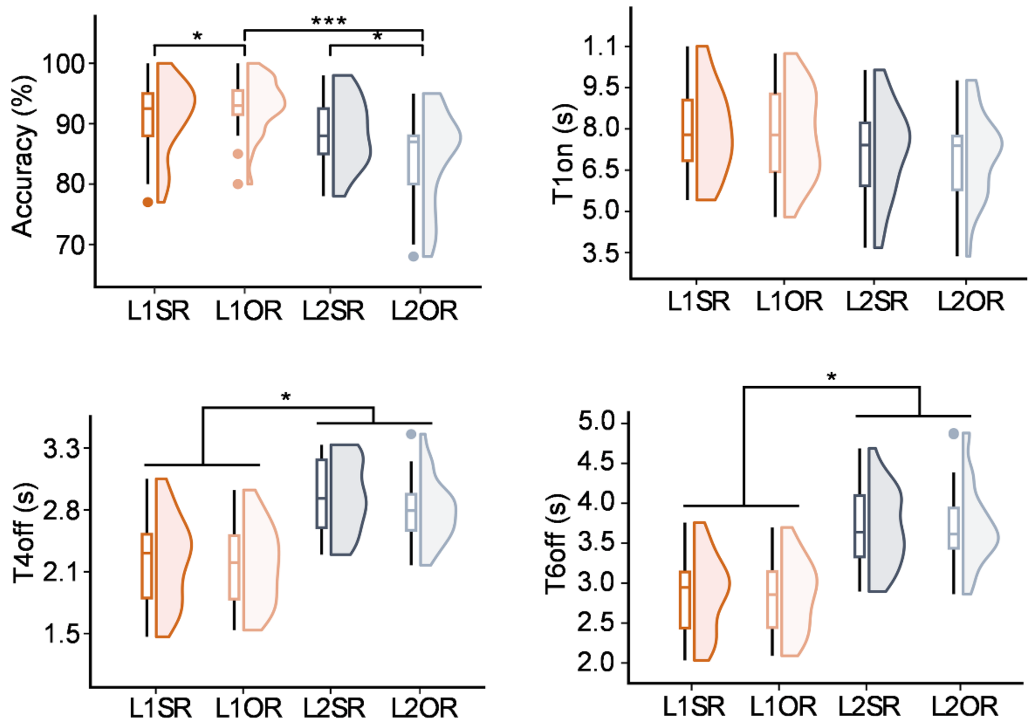

Fig S1. Behavioral performance for sentence accuracy and response times (T1on, T4off, T6off). Center lines in boxes represent the median, box limits indicate the interquartile range, and whiskers extend to the data range excluding outliers. Shaded violins show the full distribution of the behavioral data.

Table S1. Statistical results of theta power modulation in the L1 group (920-1200 ms). Results of point-by-point paired t-tests between SRC and ORC conditions. Only electrodes surviving FDR correction are listed.

Electrode	t-value	uncorrected *p*	corrected *p*
C5	-4.30	< 0.001	0.011
FC3	-3.85	< 0.001	0.011
F3	-3.77	< 0.001	0.011
FC1	-3.74	< 0.001	0.011
FC5	-3.65	0.001	0.011
P6	-3.41	0.002	0.018
CP6	-3.65	0.001	0.011
P4	-3.34	0.002	0.018

Table S2. Statistical results of theta power modulation in the L2 group (1000-3000 ms). Results of point-by-point paired t-tests between SRC and ORC conditions. Only electrodes surviving FDR correction are listed.

Electrode	t-value	uncorrected *p*	corrected *p*
F3	4.71	< 0.001	0.003
AF3	4.42	< 0.001	0.003
F5	4.41	< 0.001	0.003
Fz	3.59	0.001	0.013
F1	3.55	0.002	0.013
AF7	3.38	0.002	0.013
F2	3.37	0.002	0.013
Fpz	3.04	0.005	0.025
FC2	2.97	0.006	0.027
Fp1	2.95	0.007	0.027
AF4	2.91	0.007	0.027
F7	2.86	0.008	0.029
FT7	2.76	0.010	0.033
Fp2	2.63	0.014	0.038
FCz	2.58	0.016	0.038
FC3	2.45	0.021	0.047
CP2	3.98	0.001	0.007

Electrode	t-value	uncorrected p	corrected p
CP3	3.43	0.002	0.013
CP4	3.42	0.003	0.013
CPz	2.68	0.013	0.037
P2	3.64	0.001	0.013
P4	3.06	0.005	0.025
Pz	2.79	0.010	0.032
CP4	3.42	0.002	0.013
Pz	2.79	0.010	0.032
P3	2.61	0.015	0.038
PO7	2.6	0.015	0.038
P1	2.6	0.015	0.038
PO4	2.55	0.017	0.039

Second language proficiency

L2 proficiency was assessed using a fixed-ratio cloze test, in which Mandarin Chinese characters were systematically removed from a 301-character passage at regular intervals to create 30 gaps. L2 participants were tasked with filling these gaps with contextually appropriate Chinese characters within 15 minutes. The maximum possible score for this cloze test was 30. All L2 participants included in the analysis achieved scores of 18 or higher (with 18 representing at least intermediate Mandarin Chinese language proficiency), yielding a mean score of 25.63 ± 2.88. For a detailed description of the cloze test, see Feng et al (2020).

Materials and Methods

Stimuli. To isolate syntactic processing from other cognitive components, the lexical frequency and collocational patterns for both nouns and verbs were controlled, ensuring that participants could not rely on statistical cues unrelated to hierarchical syntactic structure. Additionally, to avoid sequence effects and prevent participants from predicting sentence structures based on prior items, two types of images were employed. This prevented participants from using directional cues (e.g., arrow trajectories) to anticipate word order.

Behavioral analysis. For all behavioral analyses, we employed mixed-effects modeling to examine the effects of sentence condition (and group, where applicable) on sentence production performance. Following Barr et al. (2013), we implemented the maximal effect structure that allowed for convergence. Specifically, in each analysis, we began with maximal models and systematically simplified the random effects structure when convergence issues arose, following the principle that the random effects structure should be kept as maximal as possible while ensuring model convergence. Below, we report the specific formulas for the final converging models used in each study.

Study 1: For native speakers, the models assessed the effect of Condition on behavioral performance. All models included random intercepts for both subjects and items.

- Accuracy: We employed a generalized linear mixed-effects model with a binomial distribution: acc ~ condition + (1|sub) + (1|item), family = binomial.
- Sentence Production Start Time (T1): t1_log ~ condition + (1|sub) + (1|item).
- Relative Clause Closure Time (T4): t4_log ~ condition + (1|sub) + (1|item).
- Overall Sentence Duration: sentence_log ~ condition + (1|sub) + (1|item).

Study 2: For the cross-group analysis involving both native speakers and L2 learners, the models assessed the effects of Condition, Group, and their interaction (Condition × Group).

- Accuracy: We employed a generalized linear mixed-effects model: acc ~ group * condition + (1|sub) + (1|item), family = binomial. The model included random intercepts for both subjects and items.
- Sentence Production Start Time (T1): t1_log ~ group * condition + (1|sub) + (1|item). This model included random intercepts for both subjects and items.
- Relative Clause Closure Time (T4): t4_log ~ group * condition + (1|sub) + (1+group|item). This model included random intercepts for both subjects and items, with an additional random slope for group by item.
- Overall Sentence Duration: sentence_log ~ group * condition + (1+condition|sub) + (1|item). This model included random intercepts for both subjects and items, with an additional random slope for condition by subject.

SI References

Barr, D. J., Levy, R., Scheepers, C., & Tily, H. J. (2013). Random effects structure for confirmatory hypothesis testing: Keep it maximal. *Journal of Memory and Language*, *68*(3), 255–278. https://doi.org/10.1016/j.jml.2012.11.001

Feng L., Feng H., Bai S., & Wu J. (2020). An Analysis of Proficiency Test for CSL (Chinese as Second Language): Based on Fixed-Ratio Cloze Questions. *Applied Linguistics*, (3), 69–79. https://doi.org/10.16499/j.cnki.1003-5397.2020.03.011